Primes congruent to -1 modulo 4
Artin, 11.13.3

Theorem 1. *There are infinitely many primes congruent to -1 modulo 4.*

Proof. We first prove the following key claim, by induction on an auxiliary bound k.

Claim. For all $k, n \in \mathbb{N}$, if $n \leq k$ and $n \bmod 4 = 3$, then there exists a prime p such that $p \mid n$ and $p \bmod 4 = 3$.

The proof is by induction on k. If $k = 0$, then $n \leq 0$, so $n = 0$, which contradicts $n \bmod 4 = 3$.

Now suppose the claim holds for k, and let $n \leq k+1$ with $n \bmod 4 = 3$. Let p be the least prime factor of n, i.e. $p = \operatorname{minFac} n$. Since $n \bmod 4 = 3$, we have $n \neq 1$, so p is prime and $p \mid n$.

The prime p is not 2: if $p = 2$, then $2 \mid n$, whereas a number divisible by 2 cannot have remainder 3 upon division by 4. Hence p is an odd prime. Since $p \bmod 4 < 4$ and p is odd, we must have

$$p \bmod 4 = 1 \qquad \text{or} \qquad p \bmod 4 = 3.$$

If $p \bmod 4 = 3$, then we are done, taking this same prime p.

Suppose instead that $p \bmod 4 = 1$. Put $m = n/p$. Since $n > 0$, as $n \bmod 4 = 3$, and since $p \geq 2$, we have $m < n$. Moreover, $p \cdot m = n$, so

$$(p \cdot m) \bmod 4 = 3.$$

But $p \bmod 4 = 1$, so $(p \cdot m) \bmod 4 = m \bmod 4$. Hence $m \bmod 4 = 3$. Also $m < n \leq k+1$, so $m \leq k$. By the induction hypothesis applied to m, there is a prime q such that $q \mid m$ and $q \bmod 4 = 3$. Since $m \mid n$, transitivity of divisibility gives $q \mid n$. This completes the induction step, and proves the claim.

Taking $k = n$ in the claim, we obtain the unbounded form: for every $n \in \mathbb{N}$ with $n \bmod 4 = 3$, there exists a prime p such that $p \mid n$ and $p \bmod 4 = 3$.

Now suppose, for contradiction, that the set

$$\{p \in \mathbb{N} \mid p \text{ is prime and } p \bmod 4 = 3\}$$

is finite. Let F be the corresponding finite set of all such primes, so that for every p,

$$p \in F \iff p \text{ is prime and } p \bmod 4 = 3.$$

We record two elementary facts about products over finite sets, both proved by induction on the finite set. First, if s is a finite set of natural numbers and every $p \in s$ is positive, then

$$0 < \prod_{p \in s} p.$$

Indeed, the empty product is 1, and inserting an element multiplies the previous product by that positive element. Second, for every finite set s and every $q \in s$,

$$q \;\Big|\; \prod_{p \in s} p.$$

Indeed, this is clear for the empty set, and when an element a is inserted, the product is a times the previous product; if $q = a$ this is immediate, and if q was already in the previous set, then q divides the previous product by induction, hence also divides a times that product.

Define

$$M = \prod_{p \in F} p.$$

Since every $p \in F$ is prime, hence positive, the first product fact gives $0 < M$.

Apply the key claim to $4M - 1$. Since $M > 0$, we have $(4M - 1) \bmod 4 = 3$. Hence there exists a prime q such that

$$q \mid 4M - 1 \qquad \text{and} \qquad q \bmod 4 = 3.$$

Therefore $q \in F$, by the characterization of F. By the second product fact, $q \mid M$, and consequently $q \mid 4M$.

Write

$$4M = qa \qquad \text{and} \qquad 4M - 1 = qb$$

for some $a, b \in \mathbb{N}$. We must have $b < a$: otherwise $a \leq b$ would imply $qa \leq qb$, i.e. $4M \leq 4M - 1$, impossible. Write $a = b + d$ for some $d \in \mathbb{N}$. Then

$$qa = qb + qd.$$

Using $qa = 4M$ and $qb = 4M - 1$, we get $qd = 1$. Thus $q \mid 1$, so $q = 1$. But q is prime, so $2 \leq q$, a contradiction.

Hence the set of primes congruent to -1 modulo 4 is not finite; that is, there are infinitely many such primes. $\square$